

Haozhe Liu

Research Scientist, Efficient AI, NVIDIA Research
haozhe.liu@kaust.edu.sa · Homepage · Google Scholar · GitHub

Updated 21 September 2026
Singapore

Research Scientist and **founding member of NVIDIA Research Singapore Lab**. As **Team Lead of SoL-Pi** and a **Core Contributor to the SANA and SoL series**, I study recursive self-improvement for agents and model inference. Before NVIDIA, I worked at Meta on multimodal foundation models and generative AI products, and earned my Ph.D. at KAUST with Jürgen Schmidhuber. My experience spans **1K+ GPU pretraining**, billion-pair datasets, and **20B-parameter models**. I have co-authored **40+ research papers** with **1.6K+ citations**, received a **Best Paper Award at the NeurIPS 2023 R0-FoMo Workshop**, and led or contributed to open-source projects with a combined **10K+ GitHub stars**.

RESEARCH EXPERIENCE

NVIDIA Research, Efficient AI

Research Scientist, working with Prof. Song Han

Singapore
2026 – present

Founding member of NVIDIA Research Singapore Lab.

- **Team Lead of SoL-Pi**. Scale auto-research loops to discover reusable harness improvements for more token-efficient coding agents.
- **Core Contributor to SoL-Series**. Advance **RSI for Model Inference** through **SoL-Engine** and **SoL-Attn**, developing workflows for inference acceleration.
- **Core Contributor to SANA-Series**. Contribute to **SANA-WM**, **SANA-Video-2.0**, and **SANA-Streaming** across world modeling, efficient video generation, and real-time streaming.

Meta, BizAI

Research Scientist Intern, with Ding Liu and Wei Liu

MPK, United States
2025

- Co-developed a commercial **7B text-to-image model**, trained on over **1B image-text pairs** and deployed in production.
- Built a **20B native LLM-driven multimodal generation prototype** from scratch.
- Contributed to models for unified visual understanding and generation, and to the **Neural Computers** project.

Meta, MGenAI

Research Scientist Intern, with Juan-Manuel Pérez-Rúa and Tao Xiang

London, United Kingdom
2024

- Co-developed a text-to-image foundation model supporting widely used products.
- Scaled autoregressive diffusion for video generation through **MarDini**.

KAUST, AI Initiative

Ph.D. Researcher, advised by Prof. Jürgen Schmidhuber

Saudi Arabia
2022 – 2026

- Studied multi-step inference in diffusion models, autoregressive models, and reinforcement learning agents.
- Developed **T-GATE** attention caching, merged into Hugging Face Diffusers, and co-developed **NLSOM** for collective intelligence among agents.

Tencent, Jarvis Lab

Research Intern, with N. He, Y. Li, and Y. Zheng

Shenzhen, China
2021 – 2022

- Conducted research on generative models and medical imaging.

RESEARCH MENTORSHIP

Tian Ye (Ph.D. candidate at HKUST; co-mentored with Dr. Enze Xie and Duomin Wang)

Research topics: Recursive self-improvement, efficient agent harnesses (SoL-Pi), and efficient video generation.

Haoyi Zhu (Ph.D. student at USTC; co-mentored with Dr. Enze Xie)

Research topics: World models and controllable video generation (SANA-WM).

Qihang Cao (Ph.D. student at NTU)

Research topics: Automated research and coding-agent harness efficiency (SoL-Pi).

Sensen Gao (Ph.D. student at NTU)

Research topics: Automated research and coding-agent harness efficiency (SoL-Pi).

Yitong Li (Ph.D. student at University of Hong Kong; co-mentored with Dr. Enze Xie)

Research topics: Model inference acceleration and agent harnesses (SoL-Engine, SoL-Pi).

Lei Zhang (Ph.D. student at UC San Diego; co-mentored with Dr. Hanrong Ye)

SELECTED PUBLICATIONS

Selected work by research direction. Full list: [Google Scholar](#).

Scalable Agentic Systems

1. **SoL-Pi: Recursively Scaling Auto-Research Loops for Efficient Agent Harness**
Haozhe Liu, T. Ye, S. Gao, et al. · **Technical report, 2026** [Paper] [Project] [Code]
2. **Mindstorms in Natural Language-Based Societies of Mind**
M. Zhuge, Haozhe Liu, F. Faccio, et al. · **Computational Visual Media, 2025**
· **Equal contribution** [Paper] [Code]
NeurIPS 2023 R0-FoMo Workshop (Oral, Best Paper) [Workshop Paper]
3. **Learning to Identify Critical States for Reinforcement Learning from Videos**
Haozhe Liu, M. Zhuge, B. Li, et al. · **ICCV 2023** [Paper] [Code]
4. **Neural Computers**
M. Zhuge, C. Zhao, Haozhe Liu, et al. · **Technical report, 2026** · **Main contributor** [Paper] [Code]

Foundation Model Development

5. **SANA-WM: Efficient Minute-Scale World Modeling with Hybrid Linear Diffusion Transformer**
H. Zhu, Haozhe Liu, Y. Zhao, et al. · **Technical report, 2026**
· **Equal contribution** [Paper] [Project] [Code]
6. **Mixture of States: Routing Token-Level Dynamics for Multimodal Generation**
Haozhe Liu, D. Liu, M. Zhuge, et al. · **CVPR 2026** [Paper] [Project]
7. **MarDini: Masked Autoregressive Diffusion for Video Generation at Scale**
Haozhe Liu, S. Liu, Z. Zhou, et al. · **TMLR 2025** [Paper] [Project]
8. **TUNA: Taming Unified Visual Representations for Native Unified Multimodal Models**
Z. Liu, W. Ren, Haozhe Liu, et al. · **CVPR 2026** · **Main contributor** [Paper] [Project]
9. **SANA-Video 2.0: Hybrid Linear Attention with Attention Residuals for Efficient Video Generation**
J. Chen, J. Yu, Y. Li, ..., Haozhe Liu, et al. · **Technical report, 2026** [Paper] [Project] [Code]
10. **SANA-Video: Efficient Video Generation with Block Linear Diffusion Transformer**
J. Chen, Y. Zhao, J. Yu, ..., Haozhe Liu, et al. · **ICLR 2026 (Oral)** [Paper] [Project] [Code]

SELECTED PUBLICATIONS (CONTINUED)

Efficient AI

11. **Faster Diffusion Through Temporal Attention Decomposition**
Haozhe Liu, W. Zhang, J. Xie, et al. · TMLR, 2025; presented at ICLR 2026 [Paper] [Code]
12. **Sol Video Inference Engine: Agent-Native Full-Stack Acceleration Framework for Efficient Video Generation**
Y. Li, J. Chen, H. Li, Haozhe Liu, et al. · Technical report, 2026
· Equal contribution [Paper] [Project] [Code]
13. **Sol-Attn: Accelerating Video Generation Inference via On-the-Fly Attention Sparsification**
H. Li, Y. Li, J. Chen, . . . , Haozhe Liu, et al. · Technical report, 2026 [Paper] [Project] [Code]
14. **Adaptive Caching for Faster Video Generation with Diffusion Transformers**
K. Kahatapitiya, Haozhe Liu, S. He, et al. · ICCV 2025 [Paper] [Project] [Code]

Earlier Research

15. **Dynamically Masked Discriminator for Generative Adversarial Networks**
W. Zhang, Haozhe Liu, B. Li, et al. · NeurIPS 2023 · Equal contribution [Paper] [Code]
16. **AdaptiveMix: Improving GAN Training via Feature Space Shrinkage**
Haozhe Liu, W. Zhang, B. Li, et al. · CVPR 2023 [Paper] [Code]
17. **Combating Mode Collapse via Offline Manifold Entropy Estimation**
Haozhe Liu, B. Li, H. Wu, et al. · AAAI 2023 [Paper] [Code]
18. **Effective Presentation Attack Detection Driven by Face Related Task**
W. Zhang, Haozhe Liu, F. Liu, et al. · ECCV 2022 · Equal contribution [Paper] [Code]
19. **Group-wise Inhibition based Feature Regularization for Robust Classification**
Haozhe Liu, H. Wu, W. Xie, et al. · ICCV 2021 [Paper] [Code]

EDUCATION

King Abdullah University of Science and Technology (KAUST)

Ph.D. in Computer Science

Advisor: Prof. Jürgen Schmidhuber

Saudi Arabia

Aug. 2022 – Mar. 2026

AWARDS & GRANTS

Best Paper Award , NeurIPS R0-FoMo Workshop, for NLSOM	2023
Co-PI , “Efficient Generative Models”, SDAIA–KAUST grant (200k SAR)	2023
4th place , ECCV NICO Challenge	2022